

Homework 1 - DS5220

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$$\begin{aligned} \textcircled{1} \text{ (a) } 3\underline{u} - 4\underline{v} &= 3[2, 7, 1]^T - 4[3, 0, 4]^T \\ &= 3 \begin{bmatrix} 2 \\ 7 \\ 1 \end{bmatrix} - 4 \begin{bmatrix} 3 \\ 0 \\ 4 \end{bmatrix} \\ &= \begin{bmatrix} 6 \\ 21 \\ 3 \end{bmatrix} - \begin{bmatrix} 12 \\ 0 \\ 16 \end{bmatrix} = \begin{bmatrix} -6 \\ 21 \\ -13 \end{bmatrix} \end{aligned}$$

$$\begin{aligned} \text{(b) } 2\underline{u} + 4\underline{v} - 5\underline{w} &= 2[2, 7, 1]^T + 4[3, 0, 4]^T - 5[1, 5, 8]^T \\ &= 2 \begin{bmatrix} 2 \\ 7 \\ 1 \end{bmatrix} + 4 \begin{bmatrix} 3 \\ 0 \\ 4 \end{bmatrix} - 5 \begin{bmatrix} 1 \\ 5 \\ 8 \end{bmatrix} \\ &= \begin{bmatrix} 4 \\ 14 \\ 2 \end{bmatrix} + \begin{bmatrix} 12 \\ 0 \\ 16 \end{bmatrix} - \begin{bmatrix} 5 \\ 25 \\ 40 \end{bmatrix} = \begin{bmatrix} 11 \\ -11 \\ -22 \end{bmatrix} \end{aligned}$$

$$\textcircled{2} \sum_{i=1}^3 c_i \underline{u}_i = [0, 3, 5]^T = \begin{bmatrix} 0 \\ 3 \\ 5 \end{bmatrix}$$

$$\Leftrightarrow c_1 \underline{u}_1 + c_2 \underline{u}_2 + c_3 \underline{u}_3 = \begin{bmatrix} 0 \\ 3 \\ 5 \end{bmatrix}$$

$$\Leftrightarrow c_1 [1, -1, 1]^T + c_2 [1, 3, 2]^T + c_3 [2, 0, 1]^T = \begin{bmatrix} 0 \\ 3 \\ 5 \end{bmatrix}$$

$$\Leftrightarrow c_1 \begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix} + c_2 \begin{bmatrix} 1 \\ 3 \\ 2 \end{bmatrix} + c_3 \begin{bmatrix} 2 \\ 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ 3 \\ 5 \end{bmatrix}$$

$$\Leftrightarrow \begin{cases} c_1 + c_2 + 2c_3 = 0 & (1) \\ -c_1 + 3c_2 = 3 & (2) \\ c_1 + 2c_2 + c_3 = 5 & (3) \end{cases}$$

In (2): $-c_1 + 3c_2 = 3 \Leftrightarrow -c_1 = 3 - 3c_2$
 $\Leftrightarrow c_1 = 3c_2 - 3$

Plug in c_1 :

$$\Delta \begin{cases} 3c_2 - 3 + c_2 + 2c_3 = 0 & (1) \\ 3 - 3c_2 + 3c_2 = 3 & (2) \\ 3c_2 - 3 + 2c_2 + c_3 = 5 & (3) \end{cases}$$

$$\Leftrightarrow \begin{cases} 4c_2 + 2c_3 - 3 = 0 & (1) \\ 3 = 3 & (2) \\ 5c_2 + c_3 = 8 & (3) \end{cases}$$

In (1): $2c_3 = 3 - 4c_2 \Leftrightarrow c_3 = \frac{3}{2} - 2c_2$

plug in c_3 inside (3):

$$5c_2 + \frac{3}{2} - 2c_2 = 8 \Leftrightarrow 3c_2 = 8 - \frac{3}{2}$$

$$\Leftrightarrow 3c_2 = \frac{13}{2}$$

$$\Leftrightarrow c_2 = \frac{13}{6}$$

Now plug in c_2 inside (1) :

$$4 \times \frac{13}{6} + 2c_3 = 3$$

$$\Leftrightarrow 2c_3 = 3 - \frac{26}{3} \Leftrightarrow 2c_3 = \frac{-17}{3}$$

$$\Leftrightarrow c_3 = \frac{-17}{6}$$

Now solve for c_1 in (1) :

$$c_1 + \frac{13}{6} + 2\left(\frac{-17}{6}\right) = 0$$

$$c_1 + \frac{-21}{6} = 0 \Leftrightarrow c_1 = \frac{21}{6}$$

$$c_1 = \frac{7}{2}$$

$$\begin{aligned} \textcircled{3} \quad \underline{u} \cdot \underline{v} &= \underline{u}^T \underline{v} = \left([2, 3, 5, 7]^T \right)^T [1, -1, 1, -1]^T \\ &= [2, 3, 5, 7] \begin{bmatrix} 1 \\ -1 \\ 1 \\ -1 \end{bmatrix} \end{aligned}$$

$$= (2)(1) + (3)(-1) + (5)(1) + (7)(-1)$$

$$= 2 - 3 + 5 - 7 = \boxed{-3}$$

④ (a) generalized cosine formula:

$$\cos \theta = \frac{a \cdot b}{\|a\| \|b\|}$$

So we need to compute $\cos \theta$ for these pairs:

$$(u, v), (u, w), (v, w)$$

* Let's start by computing the magnitudes:

$$\text{general magnitude formula: } \|a\| = \sqrt{a_1^2 + a_2^2 + a_3^2}$$

$$\hookrightarrow \|u\| = \sqrt{5^2 + 4^2 + 1^2} = \sqrt{25 + 16} = \sqrt{42}$$

$$\|v\| = \sqrt{3^2 + (-4)^2 + 1^2} = \sqrt{9 + 16 + 1} = \sqrt{26}$$

$$\|w\| = \sqrt{1^2 + (-2)^2 + 3^2} = \sqrt{1 + 4 + 9} = \sqrt{14}$$

* Now let's compute the dot products:

$$\underline{u} \cdot \underline{v} = [5, 4, 1] \begin{bmatrix} 3 \\ -4 \\ 1 \end{bmatrix} = (5)(3) + (4)(-4) + (1)(1) = 0$$

$$\underline{u} \cdot \underline{w} = [5, 4, 1] \begin{bmatrix} 1 \\ -2 \\ 3 \end{bmatrix} = (5)(1) + (4)(-2) + (1)(3) = 0$$

$$\underline{v} \cdot \underline{w} = [3, -4, 1] \begin{bmatrix} 1 \\ -2 \\ 3 \end{bmatrix} = (3)(1) + (-4)(-2) + (1)(3) = 14$$

* Finally compute the cosines for each pair:

$$\hookrightarrow \text{For } (\underline{u}, \underline{v}) \Rightarrow \cos \theta = \frac{\underline{u} \cdot \underline{v}}{\|\underline{u}\| \|\underline{v}\|} = \frac{0}{\sqrt{42} \cdot \sqrt{26}} = \boxed{0}$$

$$\text{For } (\underline{u}, \underline{w}) \Rightarrow \cos \theta = \frac{\underline{u} \cdot \underline{w}}{\|\underline{u}\| \|\underline{w}\|} = \frac{0}{\sqrt{42} \cdot \sqrt{14}} = \boxed{0}$$

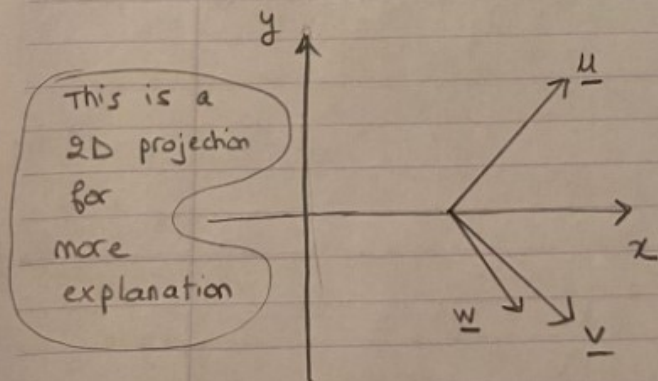
$$\text{For } (\underline{v}, \underline{w}) \Rightarrow \cos \theta = \frac{\underline{v} \cdot \underline{w}}{\|\underline{v}\| \|\underline{w}\|} = \frac{14}{\sqrt{26} \cdot \sqrt{14}} = \boxed{\frac{7}{\sqrt{91}}}$$

(b) For $(\underline{u}, \underline{v})$ and $(\underline{u}, \underline{w})$ $\cos \theta = 0$

which means they are orthogonal $\Rightarrow$ so not colinear

which leaves us with $(\underline{v}, \underline{w})$ which has

$\cos \theta = \frac{7}{\sqrt{91}} \Rightarrow$ they are the most colinear (meaning they are somewhat aligned)



so the final answer is

$$\boxed{(\underline{v}, \underline{w})}$$

(c) Since $\cos \theta = 0$ for both $(\underline{u}, \underline{v})$ and $(\underline{u}, \underline{w})$
as seen before, both are equally most orthogonal

(d) A set of vectors is linearly independent
if $c_1 \underline{v}_1 + \dots + c_k \underline{v}_k = 0$ implies $c_1 = \dots = c_k = 0$

In our case $\underline{u}, \underline{v}$ and $\underline{w}$ are linearly independent

if the only solution to $c_1 \underline{u} + c_2 \underline{v} + c_3 \underline{w} = 0$

is $c_1 = c_2 = c_3 = 0$.

We can solve this in matrix terms: $A \underline{c} = 0$

where $A = \begin{bmatrix} u_1 & v_1 & w_1 \\ u_2 & v_2 & w_2 \\ u_3 & v_3 & w_3 \end{bmatrix}$ and $\underline{c} = \begin{bmatrix} c_1 \\ c_2 \\ c_3 \end{bmatrix}$

by computing the determinant of A :

$$\det(A) = u_1 \begin{vmatrix} v_2 & w_2 \\ v_3 & w_3 \end{vmatrix} - v_1 \begin{vmatrix} u_2 & w_2 \\ u_3 & w_3 \end{vmatrix} + w_1 \begin{vmatrix} u_2 & v_2 \\ u_3 & v_3 \end{vmatrix}$$

Explanation

when we compute $\begin{vmatrix} v_2 & w_2 \\ v_3 & w_3 \end{vmatrix}$ for example, we are computing
the area of the parallelogram spanned by the
projections of $\underline{v}$ and $\underline{w}$ onto a plane perpendicular to
the x -axis. Similarly for $\begin{vmatrix} u_2 & w_2 \\ u_3 & w_3 \end{vmatrix}$ and $\begin{vmatrix} u_2 & v_2 \\ u_3 & v_3 \end{vmatrix}$

And when we multiply by u_1, v_1 and w_1 , the determinants
accumulates contributions to the volume in 3D

So basically what we are trying to do is calculate the volume of the parallelepiped spanned by $\underline{v}$, $\underline{u}$ and $\underline{w}$ in 3D. If $\det(A) \neq 0$ then the volume of this parallelepiped is NOT zero, which means that the vectors $\underline{v}$, $\underline{u}$ and $\underline{w}$ span the whole 3D space (they do NOT collapse into a lower-dimensional subspace - line or plane -). which means that they are linearly independent.

If $\det(A) = 0$ however, then the opposite occurs and the vectors are linearly dependent.

Calculation

$$\det(A) = \begin{vmatrix} 5 & 3 & 1 \\ 4 & -4 & -2 \\ 1 & 1 & 3 \end{vmatrix}$$

$$= 5 \begin{vmatrix} -4 & -2 \\ 1 & 3 \end{vmatrix} - 3 \begin{vmatrix} 4 & -2 \\ 1 & 3 \end{vmatrix} + 1 \begin{vmatrix} 4 & -4 \\ 1 & 1 \end{vmatrix}$$

$$= 5 [(-4)(3) - (-2)(1)] - 3 [(4)(3) - (-2)(1)] + [(4)(1) - (-4)(1)]$$

$$= 5 \times (-10) - 3(14) + 8 = -84$$

$\Rightarrow \det(A) \neq 0$ so the vectors are linearly independent

$$\textcircled{5} (a) * \underline{u} \cdot \underline{v} = \underline{u}^T \underline{v} = [1, 2, 1] \begin{bmatrix} 3 \\ 5 \\ 8 \end{bmatrix}$$

$$= (1)(3) + (2)(5) + (1)(8) = 3 + 10 + 8 = 21$$

$$* \|\underline{u}\| = \sqrt{1^2 + 2^2 + 1^2} = \sqrt{1+4+1} = \sqrt{6}$$

$$* \|\underline{v}\| = \sqrt{3^2 + 5^2 + 8^2} = \sqrt{9+25+64} = \sqrt{98}$$

$$* \cos \theta = \frac{\underline{u} \cdot \underline{v}}{\|\underline{u}\| \|\underline{v}\|} = \frac{21}{\sqrt{6} \cdot \sqrt{98}} = \boxed{\frac{\sqrt{3}}{2}} \approx 0.866$$

(b) general formula for projection of $\underline{u}$ onto $\underline{v}$:

$$\text{proj}_{\underline{v}} \underline{u} = \frac{\underline{u} \cdot \underline{v}}{\|\underline{v}\|^2} \underline{v}$$

$$\Rightarrow \text{proj}_{\underline{v}} \underline{u} = \frac{21}{98} \underline{v} \Leftrightarrow \frac{21}{98} \begin{bmatrix} 3 \\ 5 \\ 8 \end{bmatrix}$$

$$\Leftrightarrow \begin{bmatrix} \frac{63}{98} \\ \frac{105}{98} \\ \frac{168}{98} \end{bmatrix} \Leftrightarrow \begin{bmatrix} \frac{9}{14} \\ \frac{15}{14} \\ \frac{12}{7} \end{bmatrix} \approx \begin{bmatrix} 0.6429 \\ 1.0714 \\ 1.7143 \end{bmatrix}$$

(c) The L_2 distance (Euclidean distance) between

$$\underline{u} \text{ and } \underline{v} \text{ is } \|\underline{u} - \underline{v}\|_2 = \sqrt{(u_1 - v_1)^2 + (u_2 - v_2)^2 + (u_3 - v_3)^2}$$

$$= \sqrt{(1-3)^2 + (2-5)^2 + (1-8)^2} = \sqrt{(-2)^2 + (-3)^2 + (-7)^2}$$

$$= \sqrt{4 + 9 + 49} = \boxed{\sqrt{62}}$$

(d) The L_1 distance (Manhattan distance) is calculated using:

$$\|\underline{u} - \underline{v}\|_1 = |u_1 - v_1| + |u_2 - v_2| + |u_3 - v_3|$$

$$= |1 - 3| + |2 - 5| + |1 - 8| = 2 + 3 + 7 = \boxed{12}$$

⑥ The hyperplane equation is $(\underline{x} - \underline{p})^T \underline{w} = 0$

$$\Rightarrow \left(\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} - \begin{bmatrix} 0 \\ ? \end{bmatrix} \right)^T \begin{bmatrix} ? \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$\Rightarrow \begin{pmatrix} x_1 \\ x_2 - ? \end{pmatrix}^T \begin{bmatrix} ? \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$\Rightarrow [x_1, x_2 - ?] \begin{bmatrix} ? \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$\Rightarrow ?x_1 + (1)(x_2 - ?) = 0$$

$$\Rightarrow ?x_1 + x_2 - ? = 0 \Rightarrow \boxed{x_2 = -?x_1 + ?}$$

$$x_2 = \textcircled{m}x_1 + \textcircled{b}$$

Therefore

$$\boxed{\underline{p} = \begin{bmatrix} 0 \\ b \end{bmatrix} \text{ and } \underline{w} = \begin{bmatrix} -m \\ 1 \end{bmatrix}}$$

by comparing

⑦ To express $\underline{x}$ as linear combination of $\underline{p}_1, \underline{p}_2, \underline{p}_3$ we need to find scalars c_1, c_2 , and c_3 such

as

$$\underline{x} = c_1 \underline{p}_1 + c_2 \underline{p}_2 + c_3 \underline{p}_3$$

$$\Rightarrow \begin{bmatrix} 1 \\ 0 \\ 5 \end{bmatrix} = c_1 \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} + c_2 \begin{bmatrix} 1 \\ 1 \\ 3 \end{bmatrix} + c_3 \begin{bmatrix} 2 \\ 0 \\ -1 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} 1 \\ 0 \\ 5 \end{bmatrix} = \begin{bmatrix} c_1 + c_2 + 2c_3 \\ c_2 \\ c_1 + 3c_2 - c_3 \end{bmatrix} \quad \text{Solve for } c_1, c_2 \text{ \& } c_3$$

$$\Rightarrow \begin{cases} c_1 + c_2 + 2c_3 = 1 & (1) \\ c_2 = 0 & (2) \\ c_1 + 3c_2 - c_3 = 5 & (3) \end{cases}$$

in (1) substitute c_2 : $\begin{cases} c_1 + 2c_3 = 1 & (1) \end{cases}$

in (3) substitute c_2 : $\begin{cases} c_1 - c_3 = 5 & (3) \end{cases}$

Now in (1) : $c_1 = 1 - 2c_3$

substitute c_1 in (3) : $1 - 2c_3 - c_3 = 5$

$$\Rightarrow -3c_3 = 4 \Rightarrow c_3 = \frac{-4}{3}$$

Now substitute c_3 in (1) : $c_1 - \frac{8}{3} = 1$

$$c_1 = \frac{11}{3}$$

and $c_2 = 0$

So the linear combination is : $\underline{x} = \frac{11}{3} \underline{p}_1 + 0 \underline{p}_2 - \frac{4}{3} \underline{p}_3$

⑧ $\underline{u} = \begin{bmatrix} t \\ k \end{bmatrix}$ and $\underline{v} = \begin{bmatrix} 3 \\ 5 \end{bmatrix}$ orthogonal

$$\Rightarrow \underline{u} \cdot \underline{v} = 0 \Rightarrow \underline{u}^T \underline{v} = 0 \Rightarrow [t, k] \begin{bmatrix} 3 \\ 5 \end{bmatrix} = 0$$

$$\Rightarrow (3)(t) + (5)(k) = 0 \Rightarrow \underline{3t + 5k = 0}$$

We also know that $\underline{u}$ has unit length

$$\Rightarrow \|\underline{u}\| = 1 \Rightarrow \sqrt{t^2 + k^2} = 1 \Rightarrow \underline{t^2 + k^2 = 1}$$

$$\begin{cases} 3t + 5k = 0 & (1) \\ t^2 + k^2 = 1 & (2) \end{cases}$$

In (1): $3t = -5k \Rightarrow t = -\frac{5}{3}k$

Substitute in (2): $\left(-\frac{5}{3}k\right)^2 + k^2 = 1$

$$\Rightarrow \frac{25}{9}k^2 + k^2 = 1 \Rightarrow \frac{34}{9}k^2 = 1$$

$$\Rightarrow k^2 = \frac{9}{34} \Rightarrow \boxed{k = \pm \frac{3}{\sqrt{34}}}$$

Now substitute in (1): $t = -\frac{5}{3} \left(\pm \frac{3}{\sqrt{34}}\right)$

$$\Rightarrow \boxed{t = \pm \frac{5}{\sqrt{34}}}$$

⑨ To check if $\underline{x} = \begin{bmatrix} 2 \\ 3 \\ 3 \end{bmatrix}$ lies in the hyperplane we should see if it satisfies the equation.

$\underline{u}_1 = \begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix}$ is the normal vector (defines orthogonality)

and $\underline{w} = \begin{bmatrix} 2 \\ 2 \\ 2 \end{bmatrix}$ is a point on the hyperplane.

So the equation is $(\underline{x} - \underline{w})^T \underline{u}_1 = 0$

* computing $\underline{x} - \underline{w}$ $\Rightarrow \begin{bmatrix} 2 \\ 3 \\ 3 \end{bmatrix} - \begin{bmatrix} 2 \\ 2 \\ 2 \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}$

* So now the equation becomes:

$$\begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}^T \begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix} = 0$$

$$\Rightarrow [0, 1, 1] \begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix} = 0$$

$$\Rightarrow (0)(1) + (1)(-1) + (1)(1) = 0$$

$$\Rightarrow \boxed{0 = 0} \quad \text{The equation holds!}$$

So $\boxed{\text{YES}}$, $\underline{x} = \begin{bmatrix} 2 \\ 3 \\ 3 \end{bmatrix}$ lies on the hyperplane

$$(10) (a) \quad L\left(\begin{bmatrix} 1 \\ 0 \end{bmatrix}\right) = \begin{bmatrix} 3 \\ 5 \end{bmatrix}$$

$$\text{and} \quad L\left(\begin{bmatrix} 0 \\ 1 \end{bmatrix}\right) = \begin{bmatrix} -1 \\ 0 \end{bmatrix}$$

we want to find the matrix A such that

$$A = \left[L\left(\begin{bmatrix} 0 \\ 1 \end{bmatrix}\right) \quad L\left(\begin{bmatrix} 1 \\ 0 \end{bmatrix}\right) \right]$$

$$\Rightarrow A = \left[\begin{bmatrix} 3 \\ 5 \end{bmatrix} \quad \begin{bmatrix} -1 \\ 0 \end{bmatrix} \right] = \boxed{\begin{bmatrix} 3 & -1 \\ 5 & 0 \end{bmatrix}}$$

(b) we found before that $A = \begin{bmatrix} 3 & -1 \\ 5 & 0 \end{bmatrix}$

which means that if $\underline{x} = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$

$$\text{then} \quad L(\underline{x}) = A\underline{x} = \begin{bmatrix} 3 & -1 \\ 5 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

in this case $\underline{x} = \begin{bmatrix} 2 \\ 2 \end{bmatrix}$

$$\Rightarrow L\left(\begin{bmatrix} 2 \\ 2 \end{bmatrix}\right) = \begin{bmatrix} 3 & -1 \\ 5 & 0 \end{bmatrix} \begin{bmatrix} 2 \\ 2 \end{bmatrix}$$

$$= \begin{bmatrix} (3)(2) + (-1)(2) \\ (5)(2) + (0)(2) \end{bmatrix} = \begin{bmatrix} 6-2 \\ 10+0 \end{bmatrix}$$

$$\text{so} \quad \boxed{L\left(\begin{bmatrix} 2 \\ 2 \end{bmatrix}\right) = \begin{bmatrix} 4 \\ 10 \end{bmatrix}}$$

on the other hand, if $\underline{x} = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$

$$\text{then } L\left(\begin{bmatrix} 1 \\ -1 \end{bmatrix}\right) = \begin{bmatrix} 3 & -1 \\ 5 & 0 \end{bmatrix} \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

$$= \begin{bmatrix} (3)(1) + (-1)(-1) \\ (5)(1) + (0)(-1) \end{bmatrix} = \begin{bmatrix} 3+1 \\ 5+0 \end{bmatrix}$$

$$\text{So } \boxed{L\left(\begin{bmatrix} 1 \\ -1 \end{bmatrix}\right) = \begin{bmatrix} 4 \\ 5 \end{bmatrix}}$$

(c) To check if $\underline{x} = \begin{bmatrix} 2 \\ 3 \end{bmatrix}$ is an eigenvector of L
we need to verify the eigenvector equation:

$$L(\underline{x}) = \lambda \underline{x} \quad \text{where } \lambda \text{ is scalar} \\ \text{(the eigenvalue)}$$

$$\begin{aligned} L(\underline{x}) &= A\underline{x} = \begin{bmatrix} 3 & -1 \\ 5 & 0 \end{bmatrix} \begin{bmatrix} 2 \\ 3 \end{bmatrix} = \begin{bmatrix} (3)(2) + (-1)(3) \\ (5)(2) + (0)(3) \end{bmatrix} \\ &= \begin{bmatrix} 6-3 \\ 10+0 \end{bmatrix} = \begin{bmatrix} 3 \\ 10 \end{bmatrix} \end{aligned}$$

Now $\Rightarrow L(\underline{x}) = \begin{bmatrix} 3 \\ 10 \end{bmatrix}$ so if we go back to the

eigenvector equation: $\begin{bmatrix} 3 \\ 10 \end{bmatrix} = \lambda \begin{bmatrix} 2 \\ 3 \end{bmatrix}$

$$\Rightarrow \begin{cases} 3 = \lambda 2 \Rightarrow \lambda = \frac{2}{3} \\ 10 = \lambda 3 \Rightarrow \lambda = \frac{10}{3} \end{cases} \neq$$

So NO $\begin{bmatrix} 2, 3 \end{bmatrix}^T$
is NOT
eigenvector of L

(11) $\underline{v} = \begin{bmatrix} 2 \\ 3 \end{bmatrix}$ and $M = \begin{bmatrix} 3 & -2 \\ -2 & 3 \end{bmatrix}$

we need to verify that $M\underline{v} = \lambda \underline{v}$ (where λ is eigenvalue)

$$M\underline{v} = \begin{bmatrix} 3 & -2 \\ -2 & 3 \end{bmatrix} \begin{bmatrix} 2 \\ 3 \end{bmatrix}$$

$$\Rightarrow M\underline{v} = \begin{bmatrix} (3)(2) + (-2)(3) \\ (-2)(2) + (3)(3) \end{bmatrix} = \begin{bmatrix} 6-6 \\ -4+9 \end{bmatrix} = \begin{bmatrix} 0 \\ 5 \end{bmatrix}$$

Now check if λ exist: $\begin{bmatrix} 0 \\ 5 \end{bmatrix} = \lambda \begin{bmatrix} 2 \\ 3 \end{bmatrix}$

$$\Rightarrow \begin{cases} 2\lambda = 0 \\ 3\lambda = 5 \end{cases} \Rightarrow \begin{cases} \lambda = 0 \\ \lambda = \frac{5}{3} \end{cases} \nrightarrow \neq$$

The value of λ is inconsistent so NO, $\begin{bmatrix} 2 \\ 3 \end{bmatrix}$

is NOT an eigenvector of M

(12)(a) To verify that V is a basis for $\mathbb{R}^2$, the two vectors $\underline{v}_1 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$ & $\underline{v}_2 = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$ must satisfy:

* Linear independence

* Span (the vectors must span $\mathbb{R}^2$)

$\hookrightarrow$ check linear independence let $M = [\underline{v}_1 \ \underline{v}_2] = \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$

$$\det(M) = (1)(-1) - (1)(1) = -1 - 1 = -2$$

Since $\det(M) \neq 0$ then $\underline{v}_1$ & $\underline{v}_2$ are linearly independent.

↳ check span since v_1 & v_2 are linearly independent and $\mathbb{R}^2$ is 2D, the vectors automatically span $\mathbb{R}^2$.

Conclusion $V = \left\{ \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ -1 \end{bmatrix} \right\}$ is a basis of $\mathbb{R}^2$.

(b) we want to find K such that $[K]_V = P^{-1}BP$.

where B : matrix representation of the linear operator in the standard basis

and P : change-of-basis matrix whose columns are v_1 and v_2

$$P = [v_1 \ v_2] = \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$$

Find P^{-1} : first let's find $\det(P)$

$$\det(P) = (1)(-1) - (1)(1) = -2$$

$$\text{then } P^{-1} = \frac{1}{\det(P)} \begin{bmatrix} -1 & -1 \\ -1 & 1 \end{bmatrix} = \frac{1}{-2} \begin{bmatrix} -1 & -1 \\ -1 & 1 \end{bmatrix}$$

$$P^{-1} = \begin{bmatrix} \frac{1}{2} & \frac{1}{2} \\ \frac{1}{2} & -\frac{1}{2} \end{bmatrix}$$

Computing $K = P^{-1}BP$: $B = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$

$$P = \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} \text{ and } P^{-1} = \begin{bmatrix} \frac{1}{2} & \frac{1}{2} \\ \frac{1}{2} & -\frac{1}{2} \end{bmatrix}$$

$$\begin{aligned}
 \text{let's calculate } BP &= \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} \\
 &= \begin{bmatrix} (1)(1) + (2)(1) & (1)(1) + (2)(-1) \\ (3)(1) + (4)(1) & (3)(1) + (4)(-1) \end{bmatrix} \\
 &= \begin{bmatrix} 3 & -1 \\ 7 & -1 \end{bmatrix}
 \end{aligned}$$

$$\begin{aligned}
 \text{Now } P^{-1}(BP) &= \begin{bmatrix} \frac{1}{2} & \frac{1}{2} \\ \frac{1}{2} & -\frac{1}{2} \end{bmatrix} \begin{bmatrix} 3 & -1 \\ 7 & -1 \end{bmatrix} \\
 &= \begin{bmatrix} \frac{1}{2}(3) + \frac{1}{2}(7) & \frac{1}{2}(-1) + \frac{1}{2}(-1) \\ \frac{1}{2}(3) - \frac{1}{2}(7) & \frac{1}{2}(-1) - \frac{1}{2}(-1) \end{bmatrix}
 \end{aligned}$$

$$\Rightarrow K = \begin{bmatrix} \frac{10}{2} & -1 \\ -\frac{4}{2} & 0 \end{bmatrix} \Rightarrow K = \boxed{\begin{bmatrix} 5 & -1 \\ -2 & 0 \end{bmatrix}}$$

⑬(a) Formula for expected value : $E[Y] = A E[X]$

$$\begin{aligned}
 \Rightarrow E[Y] &= \begin{bmatrix} -1 & -2 \\ 2 & 3 \end{bmatrix} \begin{bmatrix} 2 \\ 3 \end{bmatrix} = A m_x \\
 &= \begin{bmatrix} (-1)(2) + (-2)(3) \\ (2)(2) + (3)(3) \end{bmatrix} = \begin{bmatrix} -2-6 \\ 4+9 \end{bmatrix} = \begin{bmatrix} -8 \\ 13 \end{bmatrix}
 \end{aligned}$$

$$\text{So } \boxed{E[Y] = \begin{bmatrix} -8 \\ 13 \end{bmatrix}}$$

(b) $Y = AX$, the covariance of Y is :

$$C_Y = AC_X A^T$$

We already know that $A = \begin{bmatrix} -1 & -2 \\ 2 & 3 \end{bmatrix}$ & $C_X = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$

$$\& A^T = \begin{bmatrix} -1 & 2 \\ -2 & 3 \end{bmatrix}$$

Since $C_X = I$ (identity matrix) : $AC_X = A \cdot I = A$

$$\text{So now : } C_Y = AA^T = \begin{bmatrix} -1 & -2 \\ 2 & 3 \end{bmatrix} \begin{bmatrix} -1 & 2 \\ -2 & 3 \end{bmatrix}$$

$$C_Y = \begin{bmatrix} (-1)(-1) + (-2)(-2) & (-1)(2) + (-2)(3) \\ (2)(-1) + (3)(-2) & (2)(2) + (3)(3) \end{bmatrix}$$

$$C_Y = \begin{bmatrix} 5 & -8 \\ -8 & 13 \end{bmatrix}$$

(c) The PDF of a Gaussian vector Y is given by :

$$f_Y(y) = \frac{1}{\sqrt{(2\pi)^n \det(C_Y)}} \exp\left(-\frac{1}{2}(y - E[Y])^T C_Y^{-1} (y - E[Y])\right)$$

We have $E(Y) = \begin{bmatrix} -8 \\ 13 \end{bmatrix}$; $C_Y = \begin{bmatrix} 5 & -8 \\ -8 & 13 \end{bmatrix}$; $n=2$
(dimensions of Y)

* Compute $\det(C_Y)$: $\det(C_Y) = (5)(13) - (-8)(-8) = 65 - 64 = \underline{\underline{1}}$

* Compute C_Y^{-1}

$$C_Y^{-1} = \frac{1}{\det(C_Y)} \begin{bmatrix} 13 & 8 \\ 8 & 5 \end{bmatrix}$$
$$= \frac{1}{1} \begin{bmatrix} 13 & 8 \\ 8 & 5 \end{bmatrix} = \begin{bmatrix} 13 & 8 \\ 8 & 5 \end{bmatrix}$$

* So Now the Bivariate PDF :

$$f_Y(y) = \frac{1}{\sqrt{(2\pi)^2 \cdot 1}} \exp\left(-\frac{1}{2} \left(y - \begin{bmatrix} -8 \\ 13 \end{bmatrix}\right)^T \begin{bmatrix} 13 & 8 \\ 8 & 5 \end{bmatrix} \left(y - \begin{bmatrix} -8 \\ 13 \end{bmatrix}\right)\right)$$

$$\Rightarrow f_Y(y) = \frac{1}{2\pi} \cdot \exp\left(-\frac{1}{2} \left(y - \begin{bmatrix} -8 \\ 13 \end{bmatrix}\right)^T \begin{bmatrix} 13 & 8 \\ 8 & 5 \end{bmatrix} \left(y - \begin{bmatrix} -8 \\ 13 \end{bmatrix}\right)\right)$$

(14) (a) The total probability of a joint PDF must equal 1

$$\int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f_{XY}(x, y) dx dy = 1$$

in our case

$$\int_{-4}^4 \int_2^4 f_{XY}(x, y) dy dx = 1$$

$$\Rightarrow \int_{-4}^4 \int_2^4 \frac{k}{16} dy dx = 1$$

* Integrate with respect to y first:

$$\int_2^4 \frac{k}{16} dy = \frac{k}{16} \int_2^4 1 dy = \frac{k}{16} [y]_2^4 = \frac{k}{16} (4-2) \\ = \left(\frac{k}{8} \right)$$

* Now with respect to x:

$$\int_{-4}^4 \frac{k}{8} dx = \frac{k}{8} \int_{-4}^4 1 dx = \frac{k}{8} [x]_{-4}^4 \\ = \frac{k}{8} (4 - (-4)) = \frac{k}{8} \cdot 8 = (k)$$

Therefore

$$\boxed{k = 1}$$

(b) We need to check covariance:

$$\text{Cov}(X, Y) = E[XY] - E[X]E[Y]$$

if $\text{Cov}(X, Y) = 0$ then X & Y orthogonal

* Compute $E[X]$: $E[X] = \int_{-4}^4 \int_2^4 x \cdot \frac{1}{16} dy dx$

$$= \int_{-4}^4 x \cdot \frac{1}{16} \left(\int_2^4 1 dy \right) dx$$

the y-integral : $\int_2^4 1 dy = [y]_2^4 = 4-2=2$

So now substitute: $E[X] = \int_{-4}^4 x \cdot \frac{1}{16} \cdot 2 dx$

$$= \frac{1}{8} \int_{-4}^4 x dx$$

x - integral:

$$\int_{-4}^4 x dx = \left[\frac{x^2}{2} \right]_{-4}^4 = \frac{(4)^2}{2} - \frac{(-4)^2}{2} = 0$$

Therefore $\boxed{E[X] = 0}$

* Compute $E[Y]$: $E[Y] = \int_{-4}^4 \int_2^4 y \cdot \frac{1}{16} dy dx$

$$= \int_{-4}^4 \frac{1}{16} \left(\int_2^4 y dy \right) dx$$

y - integral: $\int_2^4 y dy = \left[\frac{y^2}{2} \right]_2^4 = \frac{4^2}{2} - \frac{2^2}{2} = 6$

substitute back: $E[Y] = \int_{-4}^4 \frac{1}{16} \cdot 6 dx = \frac{6}{16} \int_{-4}^4 1 dx$

x - integral: $\int_{-4}^4 1 dx = 4 - (-4) = 8$

Therefore $\boxed{E[Y] = \frac{6}{16} \cdot 8 = 3}$

* Compute $E[XY]$: $E[XY] = \int_{-4}^4 \int_2^4 xy \frac{1}{16} dy dx$

$$= \int_{-4}^4 x \cdot \frac{1}{16} \left(\int_2^4 y dy \right) dx$$

$\rightarrow = 6$ (computed before)

$$\Rightarrow E[XY] = \int_{-4}^4 x \frac{1}{16} \cdot 6 dx = \frac{6}{16} \int_{-4}^4 x dx = 0$$

Therefore $E[XY] = 0$

* Finally $\text{Cov}(X, Y) = 0 - (0)(3) = 0$

$\Rightarrow X$ and Y orthogonal

(c) To prove that X and Y are independent, we need to show that $\rho_{X,Y}(x,y) = \rho_X(x) \rho_Y(y)$

* Step 1 $\rho_X(x) = \int_2^4 \frac{1}{16} dy$ for $-4 \leq x \leq 4$

$$= \frac{1}{16} \int_2^4 1 dy = \frac{1}{16} [y]_2^4 = \frac{1}{16} (4-2) = \frac{2}{16} = \frac{1}{8}$$

for x outside of $-4 \leq x \leq 4$, $\rho_X(x) = 0$

$$\Rightarrow \rho_X(x) = \begin{cases} \frac{1}{8} & \text{if } -4 \leq x \leq 4 \\ 0 & \text{otherwise} \end{cases}$$

$$\begin{aligned}
 * \text{ Step 2 : } f_Y(y) &= \int_{-4}^4 \frac{1}{16} dx \quad \text{for } 2 \leq y \leq 4 \\
 &= \frac{1}{16} \int_{-4}^4 1 dx = \frac{1}{16} [x]_{-4}^4 = \frac{1}{16} (4 - (-4)) = \frac{1}{2}
 \end{aligned}$$

$$\Rightarrow f_Y(y) = \begin{cases} \frac{1}{2} & \text{if } 2 \leq y \leq 4 \\ 0 & \text{otherwise} \end{cases}$$

$$* \text{ So now : } f_{XY}(x, y) = \begin{cases} \frac{1}{8} \cdot \frac{1}{2} = \frac{1}{16} & \begin{matrix} -4 \leq x \leq 4 \\ 2 \leq y \leq 4 \end{matrix} \\ 0 & \text{otherwise} \end{cases}$$

$$\text{and we have } f_{XY}(x, y) = \begin{cases} \frac{1}{16} & \begin{matrix} -4 \leq x \leq 4 \\ 2 \leq y \leq 4 \end{matrix} \\ 0 & \text{otherwise} \end{cases}$$

$$\text{so } f_{XY}(x, y) = f_X(x) f_Y(y) = \frac{1}{16}$$

Therefore X and Y are independent

$$(d) \quad P[Y \leq 3 | X \geq 0] = \frac{P[Y \leq 3 \cap X \geq 0]}{P[X \geq 0]}$$

* Compute $P[X \geq 0]$:

$$\begin{aligned}
 P[X \geq 0] &= \int_0^4 f_X(x) dx = \int_0^4 \frac{1}{8} dx \\
 &= \frac{1}{8} \int_0^4 1 dx = \frac{1}{8} [x]_0^4 = \frac{1}{8} (4 - 0) = \frac{1}{2}
 \end{aligned}$$

* Compute $P[Y \leq 3 \cap X \geq 0]$

$$P[Y \leq 3 \cap X \geq 0] = \int_0^4 \int_2^3 g_{XY}(x, y) dy dx$$

we know that $g_{XY}(x, y) = \frac{1}{16}$

so $\Rightarrow$

$$= \int_0^4 \int_2^3 \frac{1}{16} dy dx$$

y-integral :

$$\int_2^3 \frac{1}{16} dy = \frac{1}{16} [y]_2^3 = \frac{1}{16} (3-2) = \frac{1}{16}$$

substitute :

$$\int_0^4 \frac{1}{16} dx$$

$$= \frac{1}{16} [x]_0^4 = \frac{1}{16} (4-0) = \frac{1}{4}$$

* Finally :

$$P[Y \leq 3 \mid X \geq 0] = \frac{\frac{1}{4}}{\frac{1}{2}} = \boxed{\frac{1}{2}}$$

$$(15) \quad (a) \quad P(C_2|H) = \frac{P(H|C_2) P(C_2)}{P(H)}$$

$$* \quad \underline{P(H)} = P(H|C_1) P(C_1) + P(H|C_2) P(C_2)$$

(law of total probability)

we know that $P(H|C_1) = \frac{1}{7}$; $P(H|C_2) = \frac{7}{8}$

$$P(C_1) = \frac{1}{3} \quad \text{and} \quad P(C_2) = \frac{2}{3}$$

$$\underline{\underline{so}} \quad P(H) = \left(\frac{1}{7}\right)\left(\frac{1}{3}\right) + \left(\frac{7}{8}\right)\left(\frac{2}{3}\right)$$

$$\boxed{P(H) = \frac{53}{84}}$$

$$* \quad P(C_2|H) = \frac{\left(\frac{7}{8}\right)\left(\frac{2}{3}\right)}{\frac{53}{84}} = \frac{\frac{14}{24}}{\frac{53}{84}} = \frac{14}{24} \times \frac{84}{53}$$

$$\Rightarrow P(C_2|H) = \frac{1176}{1272} = \frac{49}{53} \approx 0.925$$

(b) From before, we found that $P(H) = \frac{53}{84}$
 which is NOT equal to $\frac{1}{2}$

(16) The characteristic function of a random variable X is defined as $\phi_X(t) = E[e^{itx}] = \int_{-\infty}^{\infty} e^{itx} f_X(x) dx$

in our case : $\phi_X(t) = \int_0^1 e^{itx} (2x) dx$

$$= 2 \int_0^1 x e^{itx} dx$$

using integration by parts :

let $u = x$ and $dv = e^{itx} dx$

then $du = dx$, $v = \frac{e^{itx}}{it}$

$$\Rightarrow \int u dv = uv - \int v du$$

substituting : $\int x e^{itx} dx = \left[x \frac{e^{itx}}{it} \right]_0^1 - \int_0^1 \frac{e^{itx}}{it} dx$

$$* \left[x \frac{e^{itx}}{it} \right]_0^1 = \frac{1}{it} e^{it} - \frac{1}{it} e^{i(0)} = \frac{1}{it} (e^{it} - 1)$$

$$* \int_0^1 \frac{e^{itx}}{it} dx = \frac{1}{it} \int_0^1 e^{itx} dx = \frac{1}{it} \cdot \frac{e^{itx}}{it}$$

$$* \text{Finally : } \int_0^1 x e^{itx} dx = \frac{1}{it} (e^{it} - 1) - \frac{1}{(it)^2} (e^{it} - 1)$$

$$= \frac{e^{it} - 1}{it} \left(1 - \frac{1}{it}\right)$$

$$\Rightarrow \boxed{\phi_X(t) = 2 \cdot \frac{e^{it} - 1}{(it)^2} (it - 1)}$$

(17)

$$F_Y(y) = P(Y \leq y)$$

There are two cases to analyse in order to analyze how $Y \leq y$ maps to conditions on X .

* Case 1: when $x > 0$, $Y = x$

$$\text{So } Y \leq y \Rightarrow x \leq y \quad (\text{for } y > 0, \text{ since } x > 0)$$

* Case 2: when $x \leq 0$, $Y = 2x^2$

$$\text{So } Y \leq y \Rightarrow 2x^2 \leq y \Rightarrow x^2 \leq \frac{y}{2}$$

$$\Rightarrow -\sqrt{\frac{y}{2}} \leq x \leq \sqrt{\frac{y}{2}} \quad \left(\begin{array}{l} \text{for } y \geq 0 \\ \text{since} \\ Y = 2x^2 \geq 0 \end{array} \right)$$

Now let's express $F_Y(y)$ in terms of $F_X(x)$:

* Case 1: $y < 0 \Rightarrow P(Y \leq y) = 0$ (by definition, since $Y \geq 0$)

$$\Rightarrow F_Y(y) = 0 \quad \text{for } y < 0$$

* Case 2 : for $0 \leq y \leq 2$, $F_Y(y) = P(-\sqrt{\frac{y}{2}} \leq X \leq \sqrt{\frac{y}{2}})$

$$F_Y(y) = P\left(-\sqrt{\frac{y}{2}} \leq X \leq \sqrt{\frac{y}{2}}\right) \\ = F_X\left(\sqrt{\frac{y}{2}}\right) - F_X\left(-\sqrt{\frac{y}{2}}\right)$$

* Case 3 : For $y > 2$, $F_Y(y) = P(X \leq y)$

so $F_Y(y) = F_X(y)$ since $X > 0$ for $Y = X$

Finally

$$F_Y(y) = \begin{cases} 0 & \text{if } y < 0 \\ F_X\left(\sqrt{\frac{y}{2}}\right) - F_X\left(-\sqrt{\frac{y}{2}}\right) & \text{if } 0 \leq y \leq 2 \\ F_X(y) & \text{if } y > 2 \end{cases}$$

(b) Case 1 : For $x > 0$, $Y = x$ so $x = y$ (for $y > 0$)

$$\Rightarrow b_Y(y) = b_X(x) = b_X(y)$$

let $g(x) = Y$, $g'(x) = 1$ for $x > 0$

$$\Rightarrow b_Y(y) = \frac{1}{\sqrt{2\pi}} e^{-\frac{y^2}{2}} \quad \text{for } y > 0$$

* Case 2: $0 \leq y \leq 2$ (from $x \leq 0$)

in this case $y = 2x^2$, so $x = \pm \sqrt{\frac{y}{2}}$

$$b_Y(y) = \sum_{x = \pm \sqrt{\frac{y}{2}}} \frac{b_X(x)}{|g'(x)|}$$

here, $g(x) = 2x^2$ so $g'(x) = 4x$

$$\Rightarrow |g'(x)| = 4 \left| \pm \sqrt{\frac{y}{2}} \right| = 4 \sqrt{\frac{y}{2}}$$

so now we get $b_Y(y) = 2 \frac{b_X\left(\sqrt{\frac{y}{2}}\right)}{4 \sqrt{\frac{y}{2}}}$

substitute $b_X(x) = \frac{1}{\sqrt{2\pi}} e^{-x^2/2}$ and $x = \sqrt{\frac{y}{2}}$

$$\Rightarrow b_Y(y) = \frac{2}{4 \sqrt{\frac{y}{2}}} \cdot \frac{1}{\sqrt{2\pi}} e^{-\frac{(\sqrt{\frac{y}{2}})^2}{2}}$$

$$= \frac{1}{2 \sqrt{\frac{y}{2}}} \cdot \frac{1}{\sqrt{2\pi}} e^{-\frac{y}{4}}$$

$$\Rightarrow b_Y(y) = \frac{1}{\sqrt{8\pi y}} e^{-\frac{y}{4}}; 0 \leq y \leq 2$$

* Finally the PDF of Y is :

$$p_Y(y) = \begin{cases} \frac{1}{\sqrt{8\pi}y} e^{-\frac{y}{4}} & \text{if } 0 < y \leq 2 \\ \frac{1}{\sqrt{2\pi}} e^{-\frac{y^2}{2}} & \text{if } y > 0 \\ 0 & ; \text{ otherwise} \end{cases}$$